

Patrick Piper | [github](#) | [linkedin](#)

Senior software engineer at NVIDIA building distributed ML infrastructure for large multi-modal foundation models. I design and own systems across the model lifecycle: curation and enrichment pipelines feeding model training, the training and evaluation frameworks MLE teams build on, and large-scale GPU inference for corner-case mining.

NVIDIA (2022-Present, New York, NY)

Senior Software Engineer

- **Autonomous Vehicles MLOps** (Q3 2024 - Present)
 - Built dataset enrichment and release pipeline for [Alpamayo-R1](#) and other foundation models (Pavone et al., 2025), assembling pre-training and fine-tuning sample indices at **billion-sample scale** across hundreds of scenarios and corner cases.
 - Co-leading architectural redesign of dataset enrichment pipelines for fault tolerance and idempotency to hit a daily release SLA, eliminating most manual on-call interventions across distributed stages.
 - Designed and implemented CI pipeline suite enforcing daily throughput floor for dataset enrichment, spanning smoke tests to multi-hour benchmarks on Slurm-managed CPU and GPU clusters via containerized PyTorch.
 - Created self-supervised pipeline generating Chain-of-Thought annotations across **millions** of samples using local and remote LLMs, enabling MLE interpretability analysis.
 - Reduced dataset delivery time from a multi-day, manual process to a single click for bi-weekly, **multi-million** sample releases.
 - Coordinated distributed sample enrichment across multiple HPC clusters with isolated storage backends, managing data locality and pipeline consistency during training infrastructure migration and experimentation.
 - Designed and scaled a distributed GPU-accelerated PyTorch inference pipeline from a single 20s clip to **hundreds of thousands** of clips, generating 3D and 2D visualizations for corner case identification, with results uploaded via gRPC to a central metadata store, and successfully onboarded **2 Perception DNN teams** to the pipeline.
- **Autonomous Vehicles Infrastructure** (Q3 2022 - Q2 2024)
 - Designed end-to-end KPI pipelines, model factories, and training/inference/evaluation frameworks in Python for **~10 MLE teams**.
 - Automated ONNX and TensorRT artifact generation on every model change, saving **~3 engineering hours per model iteration** for 3 Perception DNN teams.
 - Developed and maintained Python SDKs for an internal GPU cluster platform for data processing and AI/ML training and validation.
 - Owned inference-to-ground-truth matching component for KPI evaluation across **5 perception DNN teams**.

Google (2020-2022, New York, NY)

Customer Solutions Engineer

- Built and scaled data pipelines powering custom dashboards for Google Ads customers, directly supporting **millions** in revenue across two sales verticals.
- Designed and implemented Python libraries used as common components across the Google Professional Services organization, including a library managing creation and multi-threaded execution of templated SQL queries, adopted by **8 teams**.
- Earned Python Readability: internal certification for mastery of Google's coding standards.
- Taught coding classes for Google employees looking to transfer to more technical roles.

Vanguard (2019-2020, Malvern, PA)

Software Developer

- Implemented robotic process automation (RPA) solutions and drove adoption of Python for integration with RPA infrastructure and machine learning applications.

Informatica (2015-2019, Remote, US)

Senior Consultant

- Led enterprise-wide deployments of data management technology across master data management (MDM) use cases for several large clients including a financial institution, a technical staffing company, and a government agency
- Built customizations to MDM load, match, and merge operations in Java; collaborated with R&D and Global Customer Support to address customer issues and patch bugs.

Publications

P. M. Piper, J. S. Vogel, M. T. Fahrenkrug, S. J. McNamee, Q. N. Pham and G. C. Lewin, "Designing an autonomous soil monitoring robot," *2015 Systems and Information Engineering Design Symposium*, 2015, pp. 137-141, doi: 10.1109/SIEDS.2015.7116962.

Education

M.S., Computer Science, Georgia Institute of Technology (2022)

- Specialization in Machine Learning

B.S., Systems Engineering, University of Virginia (2015)

- Specialization in Electro-Mechanical Systems (robotics), minor in Engineering Business